

Insurance Analytics

What comes to Customers Mind

- Whether this company is safe or not?
- Will this company give the best offer or not?
- What is the reputation of this company in the market?

Similarly, insurers can also not understand customer behavior, fraud, policy risk, and claim surety, which is mandatory before giving a policy to someone.

Introduction

Traditionally companies are just looking for what happened in the past with Descriptive analytics. But now, the industry is demanding more, such as what will happen in the future (predictive analytics) and how actions can change the outcome (Prescriptive analytics).

The Core Idea: How Insurance Works (The Analyst's Perspective):

- Risk Transfer: Individuals transfer financial risk to an insurer.
- Risk Pooling: Insurers collect premiums from many to cover the losses of a few.
- The Data Connection: Every premium, every claim, every policy detail generates data – this is our raw material for analysis.
- The Cycle: Data informs risk assessment → Risk assessment informs pricing → Premiums collected → Claims paid → More data generated.

Why do we need data analytics in the insurance industry?

- Analyze a customer's risk and determine which client is trustworthy or may cause great loss.
- Analytics create new capabilities that empower insurers to optimize every function in the insurance value chain with the help of data-driven decision-making.
- It can also detect fraud, through which the greatest frauds happen.
- Customers can use data analytics to know which insurance company gives a minimum price with suitable offers.

Types of insurance Analytics

1, Descriptive Analysis

Descriptive analytics answer the question, "What happened?". This type of analytics is by far the most commonly used by customers, providing reporting and analysis centered on past events. It helps companies understand things such as:

- In this project, descriptive analysis would involve examining historical insurance claim data to understand past trends in claim frequency, claim amounts, types of claims, customer demographics, geographical patterns, etc. This analysis provides a baseline understanding of past performance and helps identify potential areas for improvement in the marketing strategy.

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2, Diagnostic analytics: just like descriptive analytics, uses historical data to answer a question. But instead of focusing on “the what”, diagnostic analytics addresses the critical question of why an occurrence or anomaly occurred within your data. Diagnostic analytics also happen to be the most overlooked and skipped step within the analytics maturity model. Anecdotally, I see most customers attempting to go from “what happened” to “what will happen” without ever taking the time to address the “why did it happen” step. This type of analytics helps companies answer questions such as.

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- Application to the Project: For Alpha Care Insurance Solutions, diagnostic analysis would delve deeper into the descriptive findings to uncover the factors influencing claim frequency and risk levels. This could involve identifying correlations between factors such as driver demographics, vehicle types, geographical locations, and claim frequencies. Understanding these factors helps in diagnosing the root causes of high or low-risk profiles among customers.

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3, Predictive Analysis: is a form of advanced analytics that determines what is likely to happen based on historical data using machine learning.

How To Get Started with Predictive Analytics?

At the outset of any predictive analytics build, three core elements need to be established:

- Identify a problem to solve,

- Define what is you want to predict, and

- State what you will achieve by doing so.

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- predictive analysis would be used to develop models that predict the likelihood of future insurance claims based on historical patterns. Machine learning algorithms, such as regression models or decision trees, could be applied to predict which customers are likely to be low-risk and which are high-risk. These predictions inform the optimization of marketing strategies and premium adjustments for low-risk targets.

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4, Prescriptive analytics: is the fourth, and final pillar of modern analytics. Prescriptive analytics pertains to true guided analytics where your analytics is prescribing or guiding you toward a specific action to take. It is effectively the merging of descriptive, diagnostic, and predictive analytics to drive decision making.

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- Prescriptive analytics primary aim is to take the educated guess or assessment out of data analytics and streamline the decision-making process.
- For Alpha Care Insurance Solutions, prescriptive analysis would involve recommending actionable insights based on the predictive models. For example, the analysis may recommend specific marketing campaigns or pricing strategies tailored to low-risk target segments. It could also suggest personalized offers or incentives to attract new clients from these low-risk segments, such as premium discounts or additional benefits.

Keywords



1. **Premiums** : When you purchase an insurance policy, you'll be required to make regular payments, known as premiums. These payments are typically made monthly or annually and are the cost of maintaining your insurance coverage.

2. **Total Claim**: A claim is a formal request to your insurance company for coverage or reimbursement for a loss or damage. It's essential to follow the claims process outlined in your policy.

3. **No-Claims Discount**: Many insurance companies offer a no-claims discount to policyholders who haven't filed any claims within a specified period. This can lead to lower premiums over time.

4. **Deductible/Excess**: This is the amount of money you're responsible for paying out-of-pocket on a claim before your insurance coverage starts to pay. For example, if you have a \$500 deductible and your car sustains \$2,000 in damage, you'd pay the first \$500, and your insurer would cover the remaining \$1,500.

5. **Underwriting**: Underwriting is the process insurance companies use to assess the risk of insuring you or your property. It involves evaluating factors like your driving history, location, and the type of car you own to determine if the company will offer you coverage and at what premium.